

Technology News

World's first pure white LED

Energy-efficient light-emitting diodes (LEDs) are projected as the predominant lighting source of the future, replacing the incandescent bulbs as well as compact fluorescent lights (CFLs). But, LEDs do not naturally produce white light, and getting them to look like they do adds to their production cost, making them much more expensive than an average incandescent bulb or compact fluorescent lamp. Also, the quality of their light output is not that good.

Soo-Young Park at Seoul National University, South Korea, and colleagues at the University of Valencia in Spain, have engineered a molecule that combines two light-emitting materials, orange and blue, to produce a white light in the visible light spectrum. This single molecule behaves like two separate light-producing molecules—when stimulated with a voltage, it produces orange and blue light that mix to create white. A large population of the molecules reliably produces equal quantities of orange and blue light that mix to produce an even white.

The new form of LED molecule is said to be efficient, colour stable and reproducible, making it an ideal candidate for use in LED lighting.

Printed supercapacitor to power gadgets

Electronic gadgets are shrinking in size but growing in their energy demands. Since batteries store energy chemically, these are slow to recharge. By contrast, capacitors are short-term stores of electrical energy that charge almost instantaneously but hold little en-



Isaac Daniel with Blue GPS shoes

ergy. In recent years, supercapacitors have been developed that can store thousands of times as much energy as standard capacitors. Also, these can unleash large amounts of charge very quickly.

Now, a team led by George Grüner at the University of California, Los Angeles, has printed a supercapacitor for the first time. The advance will lead to 'printed' power supplies that could be useful as gadgets become thinner, lighter and even flexible.

The supercapacitor has a power density of 70 kilowatts per kilogram to allow rapid charging and discharging. The energy density is 9 watt-hours per kilogram, meaning one kilogram holds around 32 kilojoules.

To print the supercapacitor, the team sprayed carbon nanotubes onto

a plastic film—two such films act as both the device's electrodes and charge collectors. Between the two films, the team sandwiched a gel electrolyte made by mixing a water-soluble synthetic polymer with phosphoric acid and water.

Social networking shoe

IDG (Isaac Daniel Group) has developed GPS shoes, dubbed Blue GPS because of their Bluetooth capability. These shoes enable the wearer to be tracked by friends and family through their cell phones. The wearer simply inputs up to five phone numbers of family/friends into his cell phone (eventually this capability will be expanded to up to 20 people) and pushes a button on the shoe to activate a link with those on his list.



Printed supercapacitor